SOLUSI LATIHAN BOOLEAN ALGEBRA - CANONICAL & STANDARD FORM

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PERHATIAN
- Jangan baca solusi sebelum mengerjakan soal pada dokumen latihan
- Jika ada hal yang kurang jelas dari jawaban harap email ke: **herik@cs.ui.ac.id**

PART I

3-1. Use truth table to prove the following.
a. $A'\cdot B' + A\cdot B' + A\cdot B = A + B'$
b. $A\cdot B\cdot C + A'\cdot B\cdot C + A'\cdot B\cdot C' = B\cdot C + A'\cdot B$

3-2. How many literals can be derived from n variables?
a. n b. $2n$ c. n^2 d. 2^n

3-3. How many minterms can be derived from 5 variables?
a. 5 b. 10 c. 25 d. 32

3-4. Which of the following is a sum term?
a. $A\cdot B$ b. $A'\cdot B$ c. $A+B'+C$ d. $A+B\cdot C$ e. $(A+B)\cdot C$

3-5. Which of the following is a sum-of-products expression?
a. $A\cdot B$ b. $A'\cdot B$ c. $A+B'+C$ d. $A+B\cdot C$ e. All above

3-6. Which of the following statements is valid?
a. $m43 + m34 = 0$
b. $A\cdot B + A'\cdot B' = 1$
c. $A'\cdot B'\cdot D + B'\cdot C\cdot D + A\cdot C\cdot D = A'\cdot B'\cdot D + A\cdot C\cdot D$
d. $A'\cdot B + B' = A' + B$
e. None of the above

3-7. Given a Boolean function $F(A,B,C) = \Sigma m(0, 5, 7)$. Which of the following is correct?
a. $F'(A,B,C) = \Sigma m(0, 5, 7)$
b. $F(A,B,C) = \Pi M(0, 5, 7)$
c. $F(A,B,C) = \Pi M(1, 2, 3, 4, 6)$
d. $F'(A,B,C) = \Sigma m(1, 2, 3)$
e. None of the above

3-8. Given a Boolean function $F(x,y,z) = y'\cdot(x+z') + x'\cdot z$. Which of the following is correct?

a. $F(x,y,z) = \Sigma m(0, 1)$
b. $F(x,y,z) = \Sigma m(0, 1, 2, 3, 4)$
c. $F(x,y,z) = \Sigma m(0, 1, 2, 4, 5)$
d. $F(x,y,z) = \Sigma m(0, 1, 3, 4, 5)$
e. $F(x,y,z) = \Sigma m(1, 2, 3, 4, 5)$

3-9. Given a Boolean function with 6 variables a, b, c, d, e, f. What is maxterm $M60$?

a. $a'\cdot b'\cdot c'\cdot d'\cdot e\cdot f$
b. $a'+b'+c'+d'$
c. $a'+b'+c'+d'+e+f$
d. $a\cdot b\cdot c\cdot d\cdot e'\cdot f'$
e. $a+b+c+d+e'+f'$

3-10. Given a Boolean function $F(a,b,c) = \Sigma m(0, 2, 5, 6, 7)$. If a=0, b=c=1, what is the value of F?

a. 0 b. 1 c. $a'\cdot b\cdot c$ d. a' e. unknown

3-11. Given a Boolean function $F(w,x,y,z) = w\cdot x\cdot z' + x\cdot y'\cdot z + x'\cdot z$. Which of the following minterms will appear in the sum-of-minterms expression of F?

a. $w\cdot x\cdot y\cdot z$ b. $w'\cdot x'\cdot y'\cdot z'$ c. $w'\cdot x'\cdot y\cdot z'$ d. $w\cdot x\cdot y'\cdot z$ e. $w\cdot x'\cdot y\cdot z'$

3-12. Given the following truth table of a Boolean function $F(x,y,z)$, which of the expressions represents F?

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

a. $x\cdot y'\cdot z' + x'\cdot y\cdot z + x\cdot z'$
b. $x'\cdot y\cdot z + x\cdot y' + x'\cdot y'\cdot z$
c. $x\cdot y'\cdot z + x'\cdot y'\cdot z + x'\cdot y$
d. $x\cdot y\cdot z + x'\cdot y'\cdot z + x\cdot y'$
e. None above

3-13. Given a Boolean function $F(A,B,Y,Z) = \Sigma m(0, 2, 5, 6, 8, 11, 15)$.

a. Express F in the canonical SOP form.
b. Express F in the canonical POS form.
c. Express F' in the little-m notation.
d. Express F' in the big-M notation.

JAWABAN PART I:

3-1. (a)

A	B	$A'\cdot B' + A\cdot B' + A\cdot B$	$A + B'$
0	0	1	1
0	1	0	0
1	0	1	1
1	1	1	1

(b)

A	B	C	$A\cdot B\cdot C + A'\cdot B\cdot C + A'\cdot B\cdot C'$	$B\cdot C + A'\cdot B$
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

3-2. (b) 3-3. (d) 3-4. (c) 3-5. (e) 3-6. (c) 3-7. (c)

3-8. (d) 3-9. (c) 3-10. (a) 3-11. (d) 3-12. (b)

3-13. (a) $F = A'\cdot B'\cdot Y'\cdot Z' + A'\cdot B'\cdot Y\cdot Z' + A'\cdot B\cdot Y'\cdot Z + A'\cdot B\cdot Y\cdot Z' + A\cdot B'\cdot Y'\cdot Z'$
$+ A\cdot B'\cdot Y\cdot Z + A\cdot B\cdot Y\cdot Z$

(b) $F = (A+B+Y+Z')\cdot(A+B+Y'+Z')\cdot(A+B'+Y+Z)\cdot(A+B'+Y'+Z')\cdot(A'+B+Y+Z')\cdot$
$(A'+B+Y'+Z)\cdot(A'+B'+Y+Z)\cdot(A'+B'+Y+Z')\cdot(A'+B'+Y'+Z)$

(c) $F' = \Sigma m(1, 3, 4, 7, 9, 10, 12, 13, 14)$

(d) $F' = \Pi M(0, 2, 5, 6, 8, 11, 15)$

PART II

1. Manakah yang bukan **Canonical Fom**?

   Pilih salah satu:

   a. F(a,b,c) = (a + b + c')(a' + b' + c)(a' + b + c') *(Product-of-Maxterms)*

   b. F(a,b,c) = (a + b)(c + d) (standard form POS)

   c. F(x,y,z) = x + y' + z *(Product-of-Maxterms)*

   d. F(x,y,z) = xy'z + xyz' + x'y'z' *(Sum-of-Minterms)*

2. Suatu fungsi Boolean mengandung 5 variabel A, B, C, D, E. Berapakah index dari minterm A'.B.C'.D.E'

   Minterm: 0=complement; 1=non-complement

   Index = $10010_2 = (10)_{10} \rightarrow m_{10}$

3. Suatu fungsi Boolean mengandung 5 variabel A, B, C, D, E. Berapakah index dari maxterm (A + B' + C + D' + E') ?

   Maxterm: 1=complement; 0=non-complement

   Index = $01011_2 = (11)_{10} \rightarrow M_{11}$

4. Bentuk Product of Maxterms dari fungsi **F = ab' + b'c + ac** adalah:

   F = ab' + b'c + ac

   = ab'.1 + b'c.1+ac.1

   = (ab'.(c+c'))+(b'c.(a+a'))+(ac.(b+b'))

   = ab'c + ab'c' + ab'c + a'b'c + abc + ab'c

   = ab'c + ab'c' + a'b'c + abc

   Minterm, 0 = complement; 1= non complement

   F = ∑m(5,4,1,7) ; F= ΠM(0,2,3,6)

   Maxterm, 1 = complement; 0= non-complement

   F = (a+b+c).(a+b'+c).(a+b'+c').(a'+b'+c)

Pilih salah satu:

a. F = (a+b+c)(a+b+c')(a+b'+c)(a'+b'+c)

b. F = (a+b+c')(a+b'+c')(a+b'+c)(a'+b'+c)

c. F = (a+b+c)(a+b'+c')(a+b'+c)(a'+b'+c')

d. F = (a+b+c)(a+b'+c)(a+b'+c')(a'+b'+c)

5. Bentuk Sum of minterms dari fungsi **F = (a+b')(a+c)(b'+c)** adalah:

F = (a+b')(a+c)(b'+c)

= ((a+b')+(c.c'))((a+c)+(b.b'))((b'+c)+(a.a'))

= (a+b'+c)(a+b'+c')(a+b+c)(a+b'+c)(a+b'+c)(a'+b'+c)

= (a+b'+c)(a+b'+c')(a+b+c)(a'+b'+c)

= ΠM (2,3,0,6)

F = ∑m (1,4,5,7)

F = a'b'c + ab'c' + ab'c + abc

Pilih salah satu:

a. F = ab'c' + bc + a'b'c

b. F = a'b'c' + a'b'c + ab'c + abc'

c. F = abc + ab'c' + a'b'c + ab'c

d. F = ab' + a'bc + abc

6. Lengkapilah Truth Table berikut dengan mengisi 0 atau 1 untuk

F = xy + yz + x'z.

F=xy+x'z+yz.(x+x')

F=xy+x'z+xyz+x'yz

F=xy.(1+z)+x'z.(1+y)

F=xy+x'z

x	y	z	x'	xy	x'z	F
0	0	0	1	0	0	0
0	0	1	1	0	1	**1**
0	1	0	1	0	0	0
0	1	1	1	0	1	**1**
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	1	0	**1**
1	1	1	0	1	0	**1**

7. Tuliskan bentuk Product of Maxterm dari F(x,y,z) = Σm(1,3,6,7)

F(x,y,z) = ΠM(0,2,4,5)

Maxterm: 1=complement;0=non-complement

F(x,y,z) = (x+y+z).(x+y'+z).(x'+y+z).(x'+y+z')

Pilih salah satu:

a. F = (x+y'+z)(x+y'+z)(x'+y+z)(x'+y+z')

b. F = (x+y+z)(x'+y+z')(x+y'+z)(x'+y+z)

c. F = (x+y+z)(x'+y+z')(x+y'+z')(x'+y+z)

d. F = (x'+y+z)(x'+y'+z)(x+y'+z)(x'+y+z')

8. Tuliskan bentuk Product of Sum dari sum of minterms: **F = x'y'z'+xy'z'+xyz+xyz'**

Visualisasi tabel fungsi SOM (F) dan Komplemen SOM (F')

X	Y	Z	F	F'	indeks
0	0	0	1	0	m0
0	0	1	0	1	m1
0	1	0	0	1	m2
0	1	1	0	1	m3
1	0	0	1	0	m4
1	0	1	0	1	m5
1	1	0	1	0	m6
1	1	1	1	0	m7

Langkah:

1. **Tentukan fungsi SOM**

   F(x,y,z)=∑m(0,4,7,6)

2. **Tentukan komplemen fungsi SOM**

   F'(x,y,z)=∑m(1,2,3,5)

3. **F' (SOM) = F (POM)**

   F'(x,y,z)=∑m(1,2,3,5) (SOM)→ F(x,y,z)= ΠM(1,2,3,5) (POM)

   m1' = M1 dimana (x'y'z)' = x+y+z' (De Morgan Law)

   m2' = M2 dimana (x'yz')' = x+y'+z

   m3' = M3 dimana (x'yz)' = x+y'+z'

m5' = M5 dimana (xy'z)' = x'+y+z'

F(x,y,z)= ΠM(1,2,3,5) = (x+y+z').(x+y'+z).(x+y'+z').(x'+y+z')

Sederhanakan:

F(x,y,z) =(x+y+z').(x+y'+z).(x+y'+z').(x'+y+z')

=((y+z')+(x.x')).((x+y')+(z.z'))

=(y+z'). (x+y')

Pilih salah satu:

a. F = (x + y')(y + z')

b. F = (x + z)(x' + y')

c. F = (x' + z)(x' + y)

d. F = (x + y')(x' + z)

9. Tuliskan bentuk Sum of minterms dari F(x,y,z) = ΠM(0,2,4,5)

F(x,y,z)=∑m(1,3,6,7)

Minterm: 0 = complement; 1= non complement

F=x'y'z+x'yz+xyz'+xyz

Pilih salah satu:

a. F = x'yz + x'y'z' + x'y'z + xyz'

b. F = x'y'z + x'yz + xyz' + xyz

c. F = x'yz + x'yz' + xyz + x'y'z

d. F = x'yz + xyz' + xy'z + x'y'z

10. Bagaimana bentuk dual dari fungsi boolean berikut: X'.(Y+Z').(X'+Z)

Select one:

a. X.(Y'+Z).(X+Z')

b. X' + Y.Z' + X'.Z

c. X + Y'.Z + X.Z'

d. X' + (Y.Z)' + (X'.Z)'

11. Manakah bentuk paling sederhana dari fungsi boolean berikut hanya dalam 3 literal:

X'.Y'+X.Y.Z+X'.Y

X’Y’+Y.(XZ+X’)

X’Y’+Y.(X’+Z)

X’Y’+X’Y+YZ

X’(Y+Y’)+YZ

X’+YZ

Select one:

a. X.Y' + Z

b. X'+Y.Z

c. X'.Y+Z

d. X'.Z + Y

12. Tuliskan complement dari fungsi berikut:

F = (X+Z).X'.Y + X.Y.Z

((X+Z).X'.Y + X.Y.Z)’

X’Z’+X+Y’.(X’+Y’+Z’)

a. (X.Z + X' + Y) . (X + Y + Z)

b. (X'+Z').X.Y' + X'.Y'.Z'

c. (X'.Z' + X + Y') . (X' + Y' + Z')

d. ((X.Z)' + X' + Y) . (X + Y + Z)